MEMORY RECALL IN STUDENTS CATEGORICAL VS RANDOM WORDS

Group - 10

Atharva
Ankit
Kuldeep
Nishant
Aman

THE QUESTION??

- Does Categorization Really Protect Memory from Distraction, or Just We Feel Like It?

As engineering students, we deal with a lot of information every day. From long codes to tricky formulas, we're always trying to remember and understand things that aren't easy. But we don't study in perfect, quiet places. Most of the time, we're in noisy, busy hostels, or rushing to study just before deadlines.

That's exactly why we chose this experiment. We wanted to see how well we actually learn and remember things in these real-life conditions, not just in ideal situations.

FACTOR, RESPONSES AND VARIABLES

Design: RCBD (Randomized Complete Block Design)

- Factor (Treatment):
  - Categorical Words
  - Random Words
- Blocking Variable:
  - Participants (different individuals)
- Response Variable:
  - Recall Score (out of 10)

Why RCBD?

People have different memory abilities

Blocking removes this variation

Gives more accurate comparison

Experiment Setup

Participants : 16 participants

Procedure

- Show 10 words for 30 seconds
- Show 20 words (10 correct + 10 extra)
- Participant selects 10 words
- Score recorded automatically

How experiment was conducted

- Same device and environment for all participants
- Each participant performed both conditions
- Order was randomized

ID	Block	Treatment	Score (/10)
B2401	Quiet	Animals	9
B2401	Quiet	Random	7
B2402	Quiet	Random	6
B2402	Quiet	Animals	8
B2403	Noisy	Animals	8
B2403	Noisy	Random	6
B2404	Noisy	Random	5
B2404	Noisy	Animals	7
B2405	Quiet	Animals	10
B2405	Quiet	Random	8

want to try??

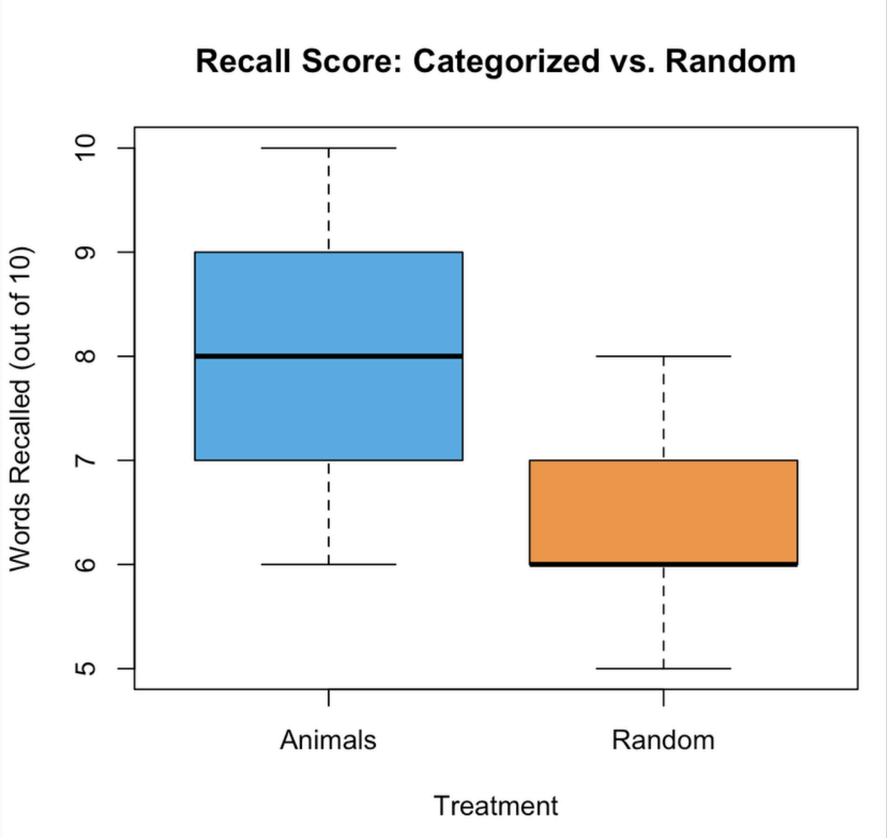
https://opixdown.github.io/Memory_recall/

ANALYSIS

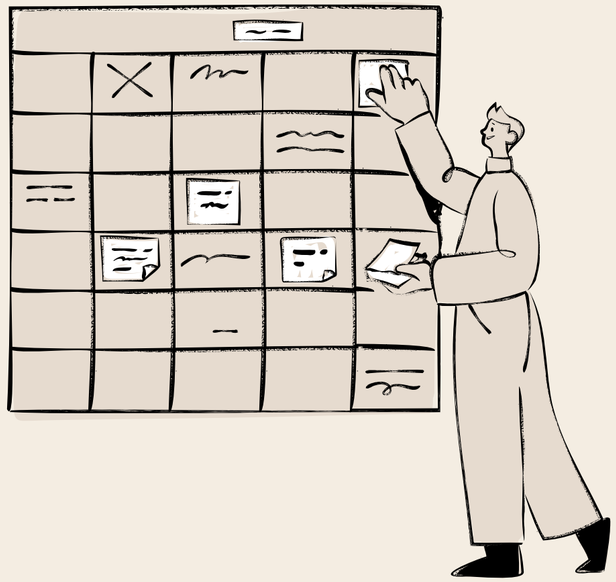
Boxplot Analysis:
The boxplot compares how many words people remembered from categorized lists (Animals) and random lists.

Key Observations:

- The categorized group has a higher median (around 8), while the random group is lower (around 6).
- People remember more words when they are organized.
- Most of the values for categorized words are higher than those for random words.
- Both groups have some variation, but categorized words still stay at higher values overall.



PLOTS & TESTS



RCBD

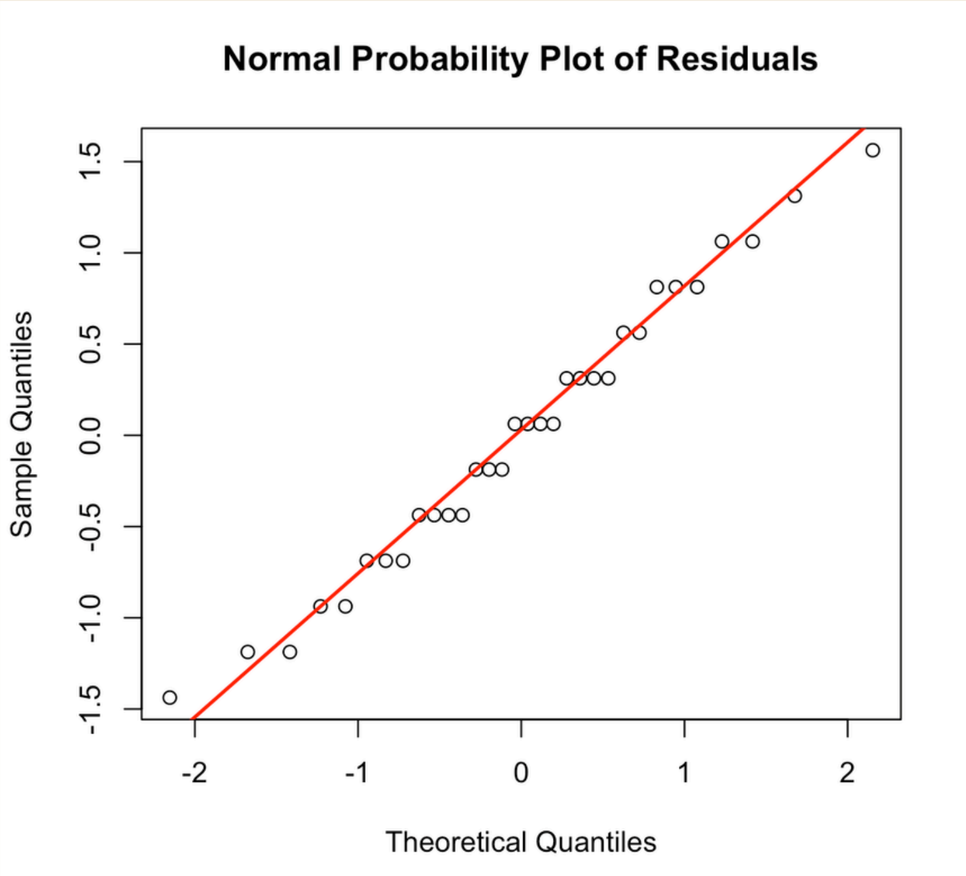
```
[1] "--- One-Way ANOVA (Ignoring Blocks) ---"
> summary(model_crd)
            Df Sum Sq Mean Sq F value   Pr(>F)
Treatment    1  24.50  24.500   20.21 9.65e-05 ***
Residuals   30  36.38   1.212
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

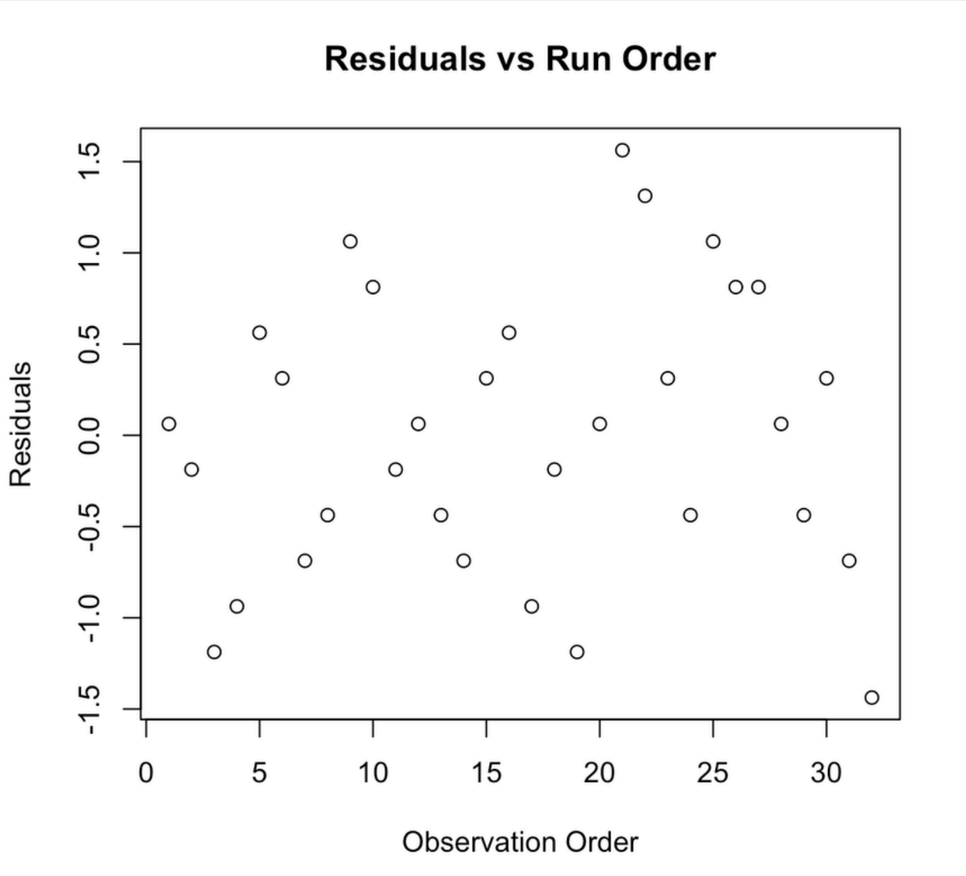
Bartlett test

```
        Bartlett test of homogeneity of variances

data:  Score by Treatment
Bartlett's K-squared = 0.22388, df = 1, p-value = 0.6361
```

PLOTS & TESTS


Normal Probability Plot of Residuals
Sample Quantiles
Theoretical Quantiles
-1.5
-1.0
-0.5
0.0
0.5
1.0
1.5
-2
-1
0
1
2


Residuals vs Run Order
Residuals
Observation Order
-1.5
-1.0
-0.5
0.0
0.5
1.0
1.5
0
5
10
15
20
25
30

ANOVA ANALYSIS

Treatment Effect – Word Type

We found a clear and strong effect of word type on memory ($F = 38.67$, $p < 0.001$).

In simple terms, people remembered more words when they were grouped into categories rather than shown randomly.

>Organized (categorical) words help improve memory.

Block Effect – Participants

There were also noticeable differences between participants ($F = 28.41$, $p < 0.001$).

This means everyone has a different memory ability.

> That's why using RCBD is important it helps control these differences.

Model Fit

The residual error is low (MS = 0.634), which means our model explains the data well.

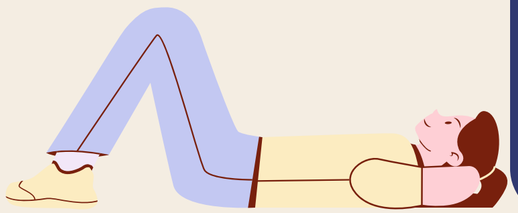
>Most of the changes in memory scores are due to our tested factors, not random noise.

The Bartlett test ($p = 0.6361$) shows that the variances are equal, meaning the ANOVA assumption of homogeneity is satisfied.

CONCLUSION

Overall, our experiment shows that the way information is organized really matters. When words are grouped into meaningful categories, people are able to remember more compared to when the same words are presented randomly.

At the same time, we also saw that memory ability varies from person to person, which is why using RCBD helped us get more accurate and fair results. In short, a little structure goes a long way in improving memory even our brains prefer things to be organized rather than chaotic... just like we prefer neatly arranged notes over last minute messy scribbles



THANK YOU

Presented by Group 10